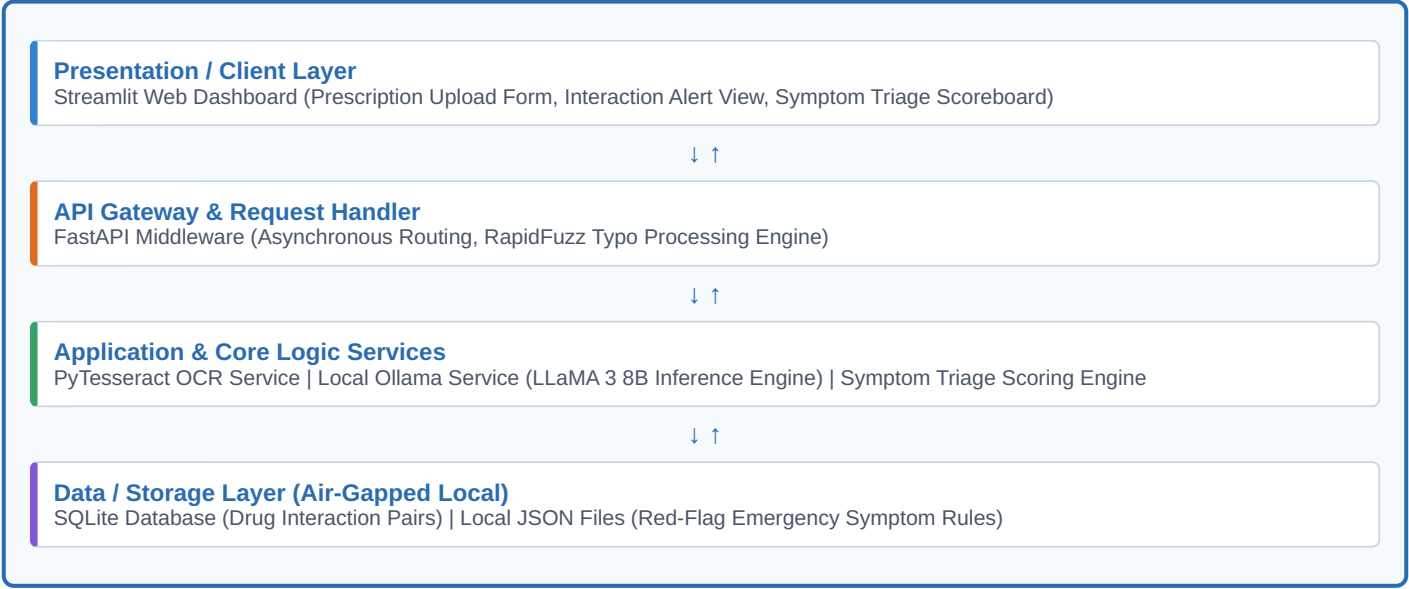


SMARTBRIDGE — Solution Architecture

System Architecture Layers & Component Details for MedSafe AI

Date:	August 12, 2026	Team ID:	MEDSAFE_AI_01
Project Name:	MedSafe AI — Healthcare Safety & Preventive Awareness Platform	Max Marks:	5 Marks

1. Visual Solution Architecture Diagram



2. Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders user interfaces for uploading prescription images, querying drug pairs, displaying triage scores, and downloading patient reports.	Streamlit (Python framework)
API Gateway	Handles incoming client requests, manages async routing to backend pipelines, and applies RapidFuzz string normalization to inputs.	FastAPI, Uvicorn
Microservices / Core Logic	Executes optical character recognition (OCR) on images, performs local LLM summarization, and calculates 1-10 symptom risk scores.	PyTesseract, Ollama (LLaMA 3 8B), Python Rule Engine
Database / Storage	Stores pre-compiled drug interaction datasets, severity mappings, and emergency red-flag keyword dictionaries offline.	SQLite 3, JSON Datasets